



WorkshopPLUS: SharePoint Server Administration

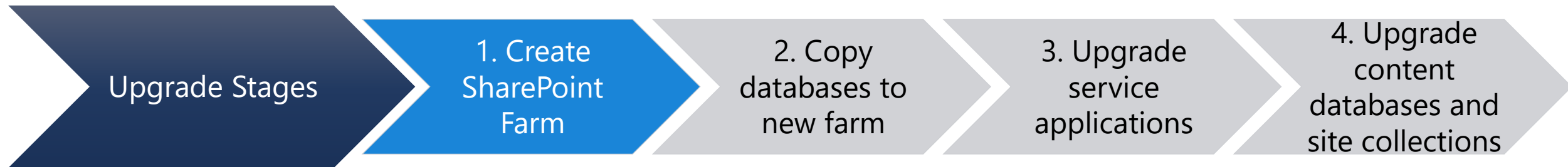
What is the Upgrade Process



Agenda - Chapter 1

1	Upgrade Versions
2	Content Databases
3	Working with Site Collections
4	Additional Considerations

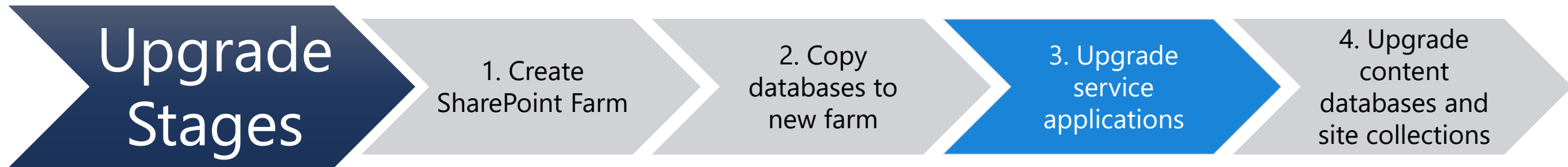
Upgrade Stages - Step 1



Upgrade Stages - Step 2



Upgrade Stages - Step 3



Upgrade Stages - Step 4



Upgradeable Versions



SharePoint 2016

Upgradeable to SharePoint 2019 and
Subscription Edition



SharePoint 2019

Upgradeable to SharePoint Subscription
Edition



All versions

Standard
Enterprise



Older Versions

Will have to upgrade to SharePoint 2016 to
use Database Attach
Third Party Tools

Agenda - Chapter 2

1

Upgrade Versions

2

Content Databases

3

Working with Site Collections

4

Additional Considerations

Content Database Upgrade

Configure Web Applications



Add AAMs



Add managed paths



Use Test-SPContentDatabase to check for errors



Use Mount-SPContentDatabase to attach database

Can upgrade multiple databases in parallel



Use Upgrade-SPContentDatabase to resume failed upgrade

Test-SPContentDatabase

Finds issues with content databases when compared to a specific web application

Watch for Upgrade Blocking

Fix Errors

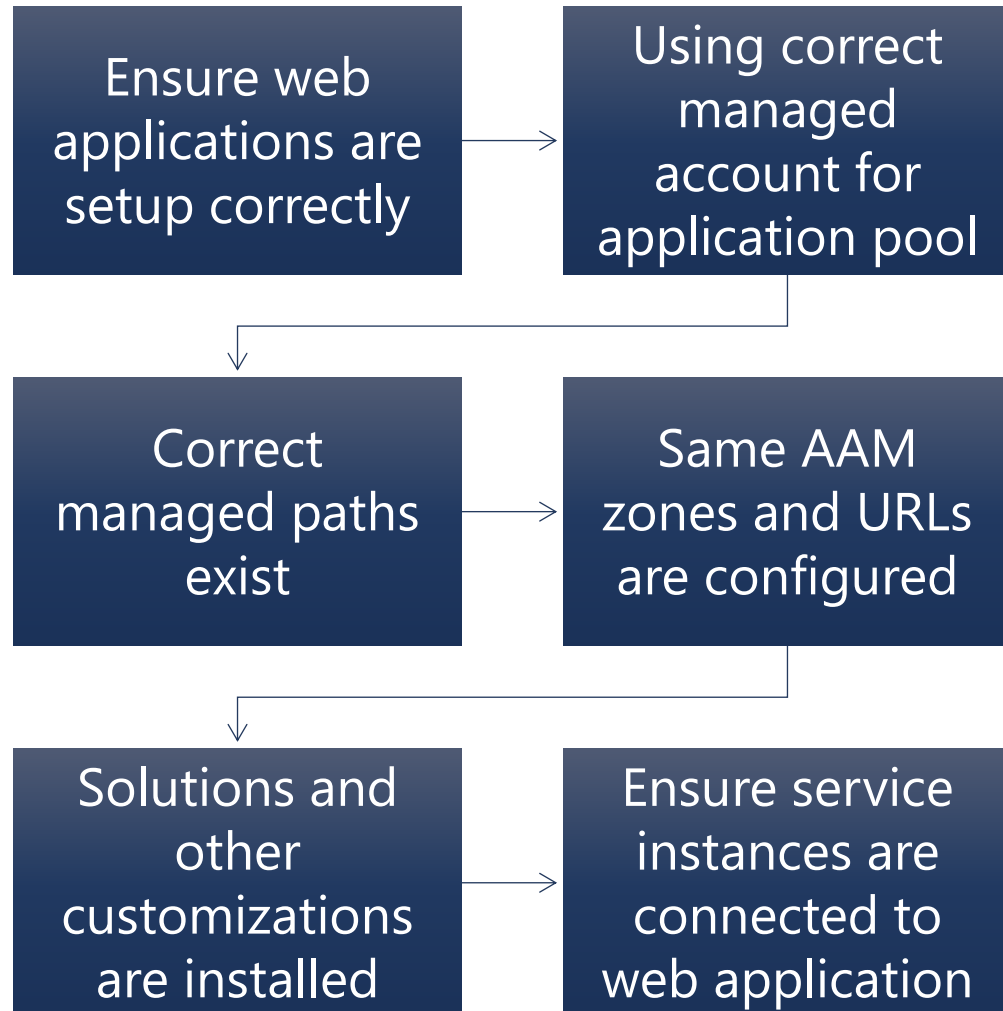
Missing Customizations

Might generate false-positive entries for upgraded features

Orphans

Web App Configuration

Before attaching content databases



- Some content database upgrades attempt to use service instances
- Service unavailability should not block upgrade
- Service issues during upgrade may require additional work afterwards
 - To finish the specific service interaction missed during upgrade
- Check log to look for issues

Demonstration

Test-SPContentDatabase



Agenda - Chapter 3

1

Upgrade Versions

2

Content Databases

3

Working with Site Collections

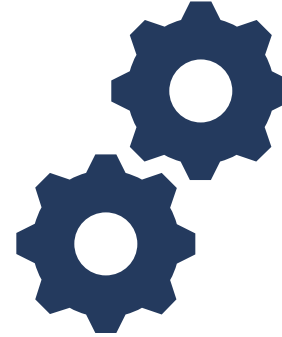
4

Additional Considerations

Site Collections



Must be in 15 mode



**Upgrade 14 mode site collections to
15 mode**

```
Get-SPSite -ContentDatabase <database name> -Limit All | Where-Object {  
    $_.CompatibilityLevel -eq 14 }
```

```
Upgrade-SPSite <url> -VersionUpgrade
```

Site Collection Upgrade



**Upgrade sites along with
database upgrades
(default in v2v upgrade).**



**Manually trigger site
upgrade (default in b2b
upgrade)**



**First browse after
database upgrade**

Verify Upgrade



Upgrade Status page

In Central Administration: Click “Upgrade and Migration” and then “Check upgrade status”



To view the upgrade log file

Located at:
%COMMONPROGRAMFILES%\Microsoft Shared\web server extensions\16\LOGS.

The logs are text files named in the following format:

- **Error:** Upgrade-YYYYMMDD-HHMMSS-SSS-error.log
- All info: Upgrade-YYYYMMDD-HHMMSS-SSS.log



Using PowerShell

```
Get-SPContentDatabase |  
Format-Table Name,  
NeedsUpgradeIncludeChildren
```

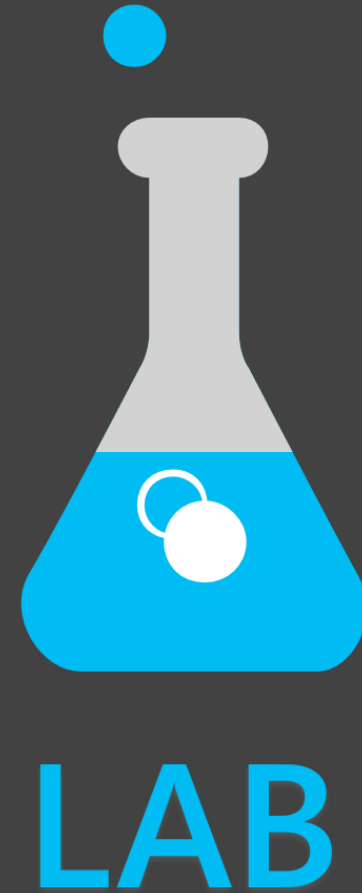
Lab 3:

Upgrade SharePoint

Objectives

After completing this lab, you will be able to:

- Run the Test-SPContentDatabase cmdlet and review errors.
- Upgrade a content database
- Check the success of an upgraded content database



Knowledge Check 1

What version of SharePoint can be directly upgraded to 2019?

- SharePoint 2016

What version of SharePoint can be directly upgraded to Subscription Edition?

- SharePoint 2016
- SharePoint 2019

Agenda - Chapter 4

1

Upgrade Versions

2

Content Databases

3

Working with Site Collections

4

Additional Considerations

Authentication Mode Support & Upgrade

Windows Classic Support (Legacy)

- Only for migration purposes, no production use
- Not possible anymore in SP Subscription Edition

Windows Claims Support

- Supported in SharePoint Server 2010 and later
- Migration before upgrade recommended

Forms Authentication Support

- No changes since SharePoint Server 2010
- Ensure provider installed with same name before database attachment

Database to Web Application authentication mode mismatches

- Database attach detects mismatched authentication support
- Test-SPContentDatabase also detects this

Claims Migration Best Practices



Migration of
authentication and
version



Find authentication
mode mismatch issues
between content
databases and web
applications with Test-
SPContentDatabase



Ensure all external data
source/web services
work as expected after
claims migration



SharePoint 2016/2019
and later do not
support classic
authentication in
production

SPSE no longer has
classic authentication

Service Application Upgrades



Collect information

Gather applicable services settings on source farm



Migrate

Backup all applicable services databases from source farm



Prepare

See steps below



Post migration

Start service instances only after service is created with upgraded databases

Preparation steps:

- Ensure service application pools are created first
- Make sure the managed account is the correct one for your security needs
- Create each service application and proxy
- If applicable, upgrade service database during service application creation
- Pay special attention to differences in some services
- Most service upgrade processes are the same, some are different though

Service Application Upgrade (Example)

```
$applicationPool = Get-SPServiceApplicationPool -Identity "SharePoint Service Application"

# Create Business Data Catalog service and proxy
$sa = New-SPBusinessDataCatalogServiceApplication -Name "Business Data Catalog Service Application" -ApplicationPool $applicationPool -DatabaseName "BDC_Upgrade_DB"
if ($null -ne $sa) {
    New-SPBusinessDataCatalogServiceApplicationProxy -ServiceApplication $sa -Name "Business Data Catalog Service Application Proxy" -DefaultProxyGroup

    #Start Business Data Catalog service
    Get-SPServiceInstance | Where-Object {$_.TypeName -Eq "Business Data Connectivity Service"} | Start-SPServiceInstance
}
```

Search Planning

Two search architectures based on core search

- Traditional Search Service Application (SSA)
- Cloud Search Service Application (CSSA)

Search experience unchanged in SharePoint 2019 and SE

- Can crawl SharePoint, Exchange, File Servers, etc.
- Localized and/or federated search results across topologies and/or cloud
- Localizes index storage

Cloud Search Service Application



New search experience since SharePoint Server 2013 January 2016 CU

Can still crawl SharePoint, Exchange, File Servers, etc.

Localized and cloud search results across topologies and/or cloud

Externalizes index storage (in Office 365)



Adds Office Graph/Delve experience

Use Search to Ease Upgrade

Cloud Search Service Application

- Ease transition from SharePoint 2013 to 2016 and 2019
 - Crawl content in Cloud Search Service Application prior to migration
- Content in one Index
 - Available after migration
- Requires that URL is not changed

Knowledge Check 2

What are the two options for search?

- Search Service Application (SSA)
- Cloud Search Service Application (Cloud SSA)

Questions?



